

Task 4: Classification Models, Evaluation Metrics & Handling Imbalanced Data

Objective

The objective of this task is to implement and evaluate classification models using the Breast Cancer Dataset. The task focuses on understanding classification metrics, ROC-AUC analysis, handling imbalanced data, and comparing different machine learning models.

Dataset

Breast Cancer Wisconsin Dataset

- Source: Scikit-learn Built-in Dataset
 - Total Samples: 569
 - Features: 30 numerical features
 - Target Classes:
 - 0 → Malignant
 - 1 → Benign
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Technologies Used

- Python
 - NumPy
 - Pandas
 - Matplotlib
 - Scikit-learn
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Concepts Covered

- Classification Models
- Logistic Regression
- Decision Tree Classifier
- Train-Test Split
- Feature Scaling
- Confusion Matrix
- Precision
- Recall
- F1-Score

- ROC Curve
 - AUC Score
 - Imbalanced Data Handling
 - Model Comparison
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Implementation Steps

1. Import Required Libraries

Imported essential libraries for data processing, visualization, model building, and evaluation.

2. Load Dataset

Loaded the Breast Cancer dataset using Scikit-learn.

3. Data Preparation

- Created feature matrix (X)
- Created target variable (y)

4. Train-Test Split

Split the dataset into:

- Training Set (80%)
- Testing Set (20%)

5. Feature Scaling

Applied StandardScaler to standardize feature values.

6. Logistic Regression

- Trained Logistic Regression model
- Generated predictions on test data

7. Model Evaluation

Evaluated model performance using:

- Confusion Matrix
- Precision
- Recall
- F1-Score
- Accuracy

8. ROC Curve & AUC

- Generated ROC Curve
- Calculated AUC Score

9. Handling Imbalanced Data

Implemented:

```
class_weight='balanced'
```

to demonstrate handling of class imbalance.

10. Decision Tree Classifier

- Trained Decision Tree model
- Evaluated performance
- Compared results with Logistic Regression

Results

Confusion Matrix

```
[[41  1]
 [ 1 71]]
```

Classification Report (Logistic Regression)

	precision	recall	f1-score	support
0	0.98	0.98	0.98	42
1	0.99	0.99	0.99	72
accuracy			0.98	114
macro avg	0.98	0.98	0.98	114
weighted avg	0.98	0.98	0.98	114

ROC-AUC Score

AUC = 1.00

Decision Tree Performance

	precision	recall	f1-score	support
0	0.85	0.93	0.89	42
1	0.96	0.90	0.93	72
accuracy			0.91	114

Model Comparison

Metric	Logistic Regression	Decision Tree
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Accuracy	98%	91%
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Metric Logistic Regression Decision Tree

Precision	98–99%	85–96%
Recall	98–99%	90–93%
F1-Score	98–99%	89–93%
AUC	1.00	Not Evaluated

Best Model

✓ Logistic Regression

Reason:

- Higher Accuracy
- Better Precision and Recall
- Better F1-Score
- Excellent ROC-AUC Score
- Only 2 Misclassifications

🔗 Key Learning Outcomes

- Understood classification algorithms.
- Implemented Logistic Regression and Decision Tree models.
- Learned how to evaluate models using Confusion Matrix and Classification Report.
- Understood Precision, Recall, and F1-Score.
- Visualized model performance using ROC Curve.
- Learned the importance of AUC Score.
- Explored techniques for handling imbalanced datasets.
- Compared multiple models and selected the best-performing classifier.

✓ Conclusion

In this task, classification models were implemented and evaluated using the Breast Cancer Dataset. Logistic Regression achieved superior performance with approximately **98% accuracy**, an **AUC score of 1.00**, and only **two misclassifications**. Based on the evaluation metrics and model comparison, Logistic Regression was selected as the final model for this classification problem.